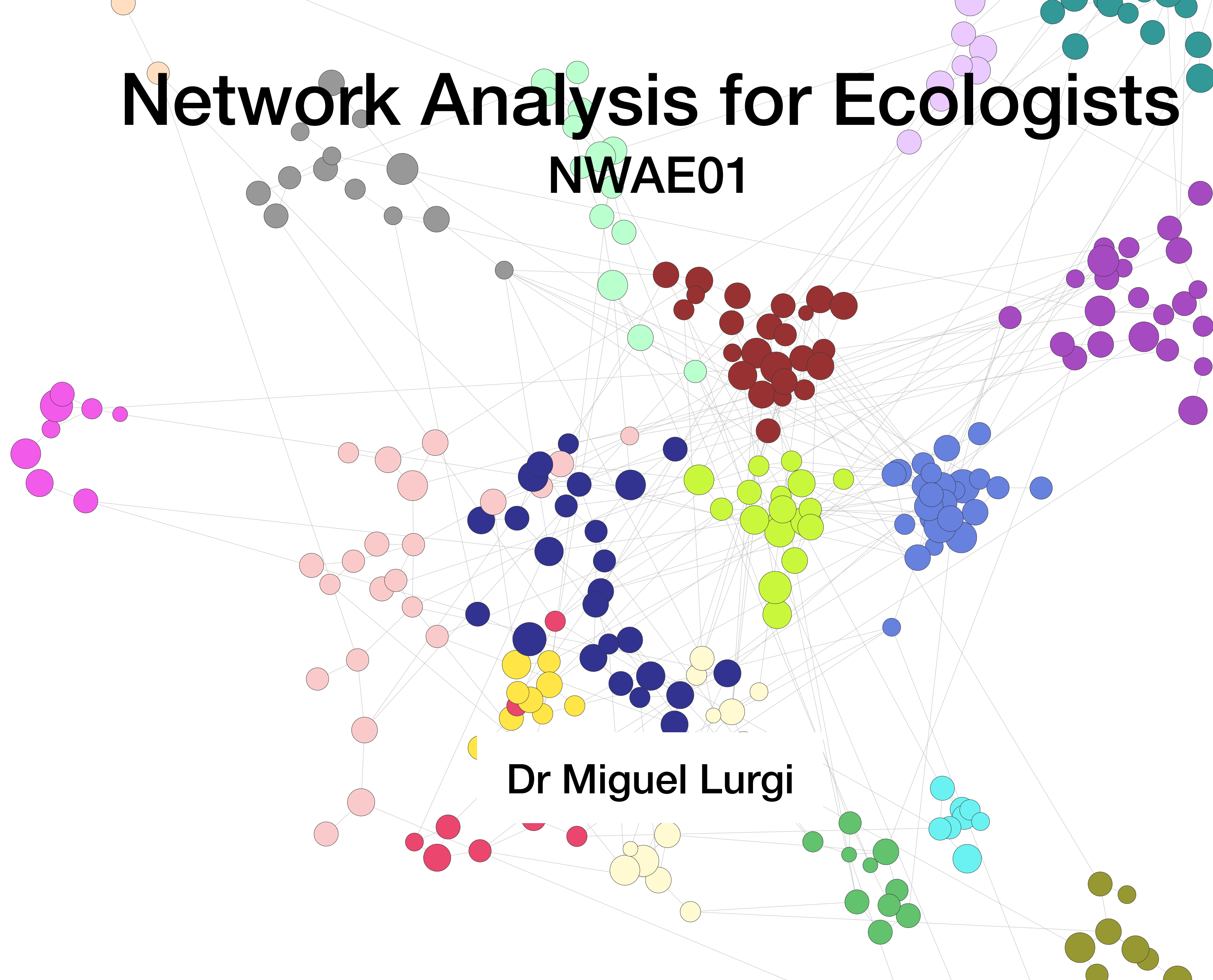


Network Analysis for Ecologists

NWAE01



Dr Miguel Lurgi

What will we learn today?

Part I : Modelling network dynamics

- Introduction to population dynamics in single and multi species systems
- Simulating communities using ecological models

Part II : Methods for data simulation

- Ordinary Differential Equations (ODEs)
- Developing an experimental protocol
- Analyse the outcomes of simulation experiments with networks

Modelling Network Dynamics

Population ecology

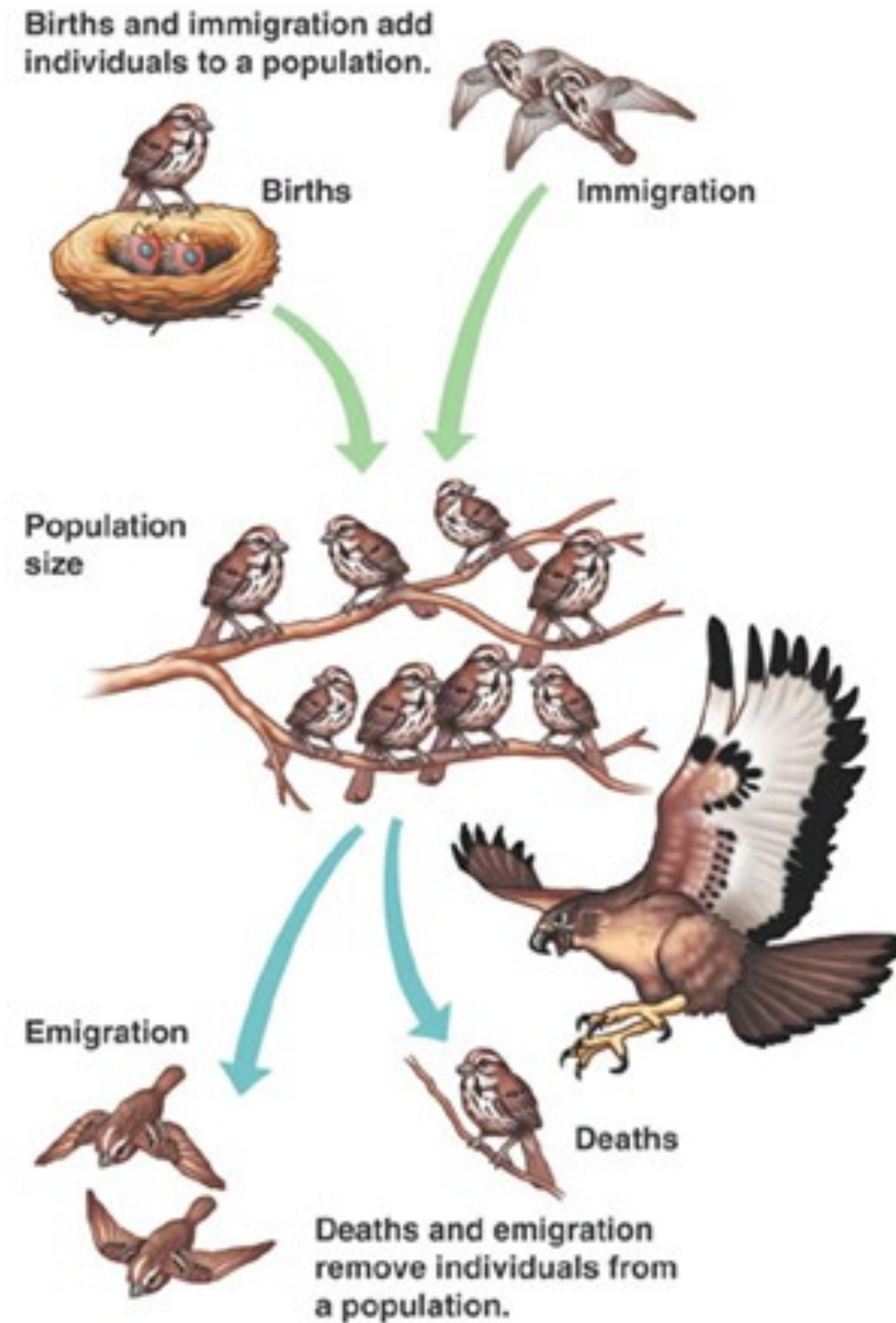
The study of changes in the size of a population (e.g. number of individuals) of a given species through time (or space) and the factors (mechanisms) that regulate it

Definition - Population: Group of individuals of a single species that occupy the same general area

This means that individuals:

- Rely on the same resources
- Are influenced by the same biotic and abiotic factors
- Have a high probability of interacting, e.g. competing, co-operating, breeding with other individuals in the group

Factors influencing growth



Limits to growth

$$N_{t+1} = N_t + B_t + I_t - D_t - E_t$$

Number of individuals

How fast it grows

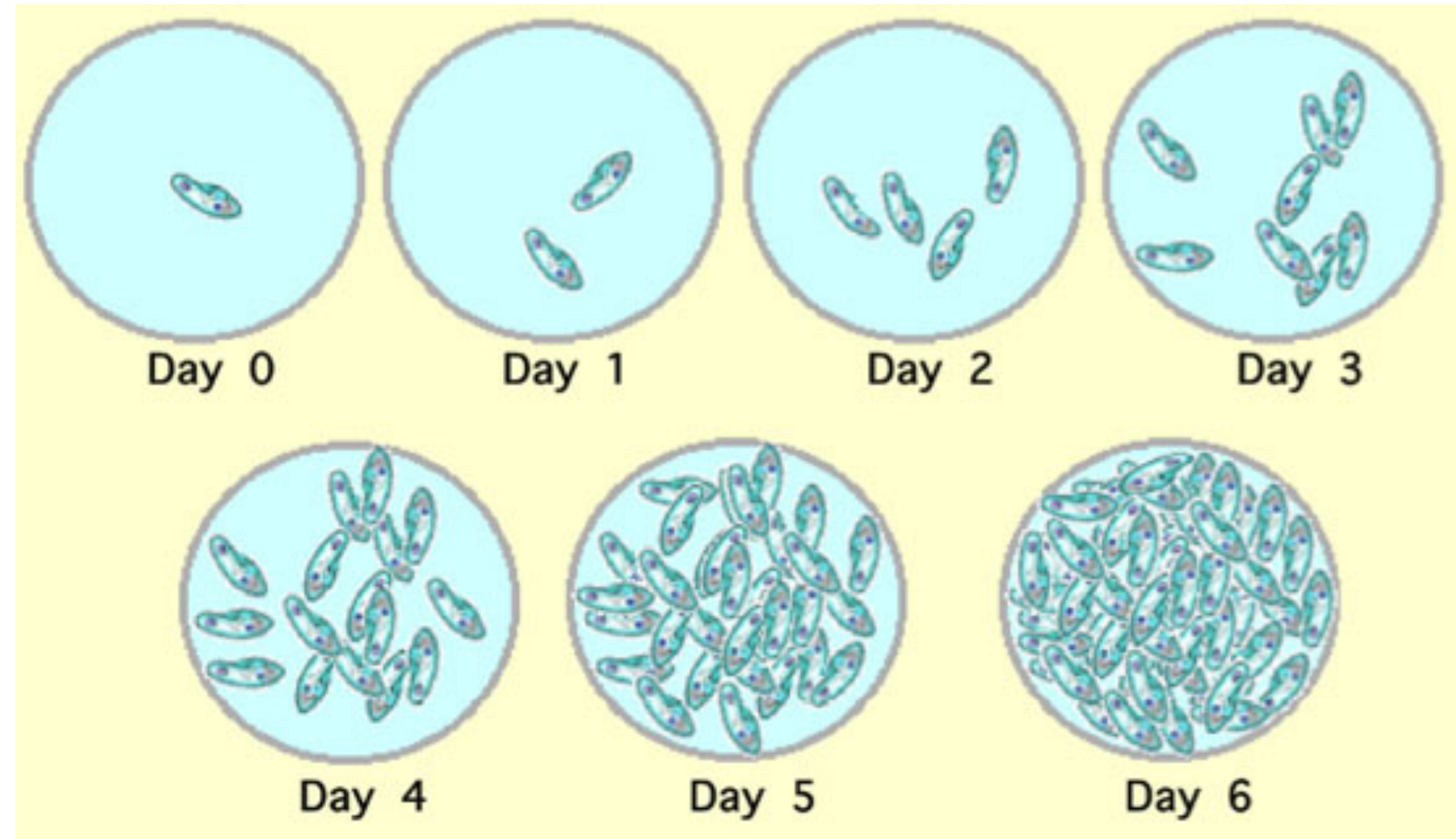
Functional forms

- Density independent growth (e.g. exponential)
- Density dependent growth

Exponential growth

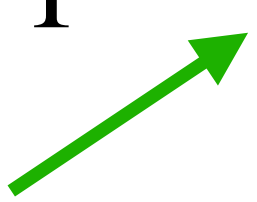
Populations growth exponentially if:

- Birth and death rates are constant
- More births than deaths
- There are no limiting resources

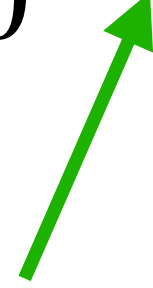


For local, closed populations

Discrete time equation

$$N_{t+1} = \textcircled{R} N_t$$


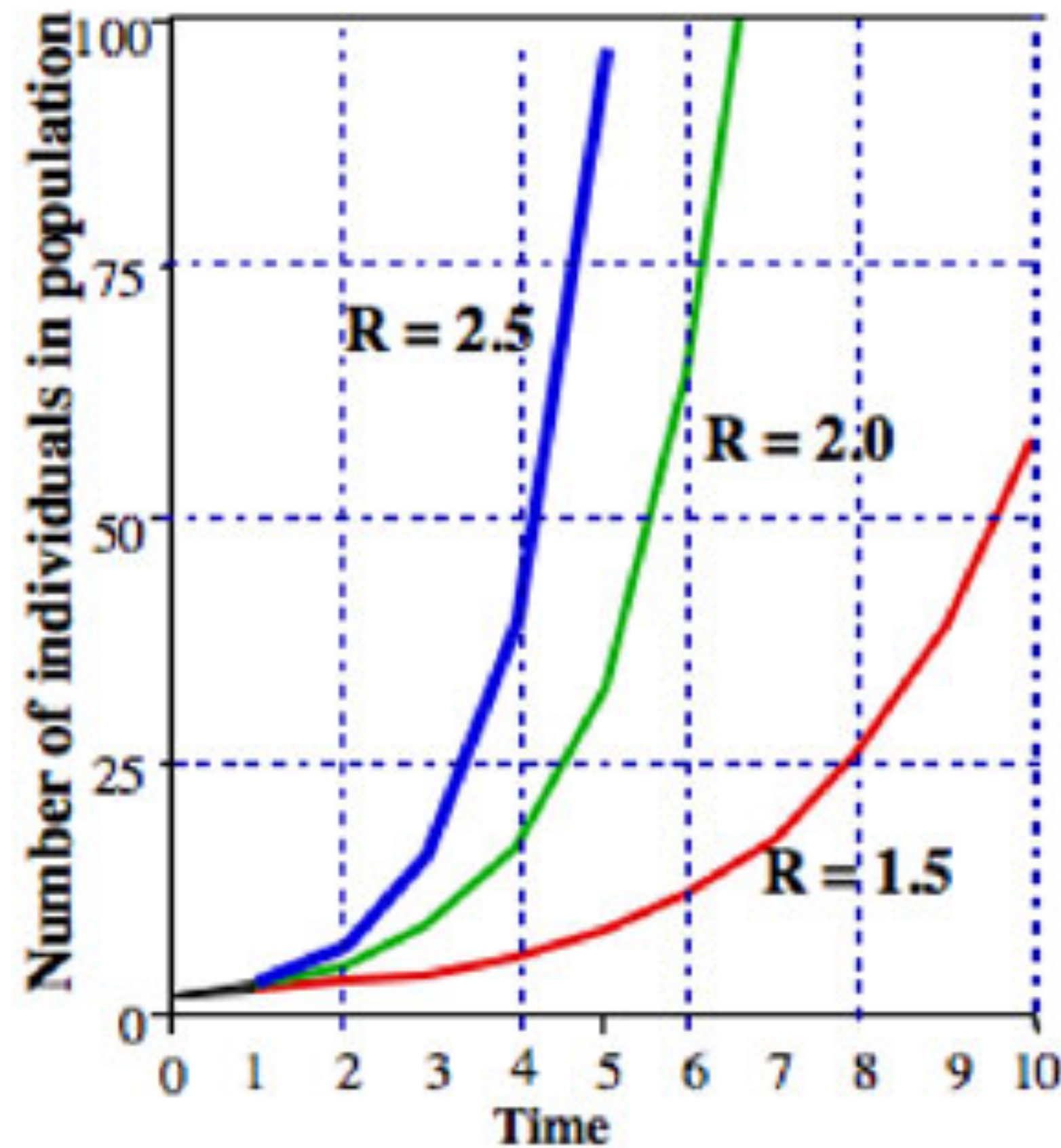
Finite rate of population increase

$$N_t = N_0 e^{\textcircled{rt}}$$


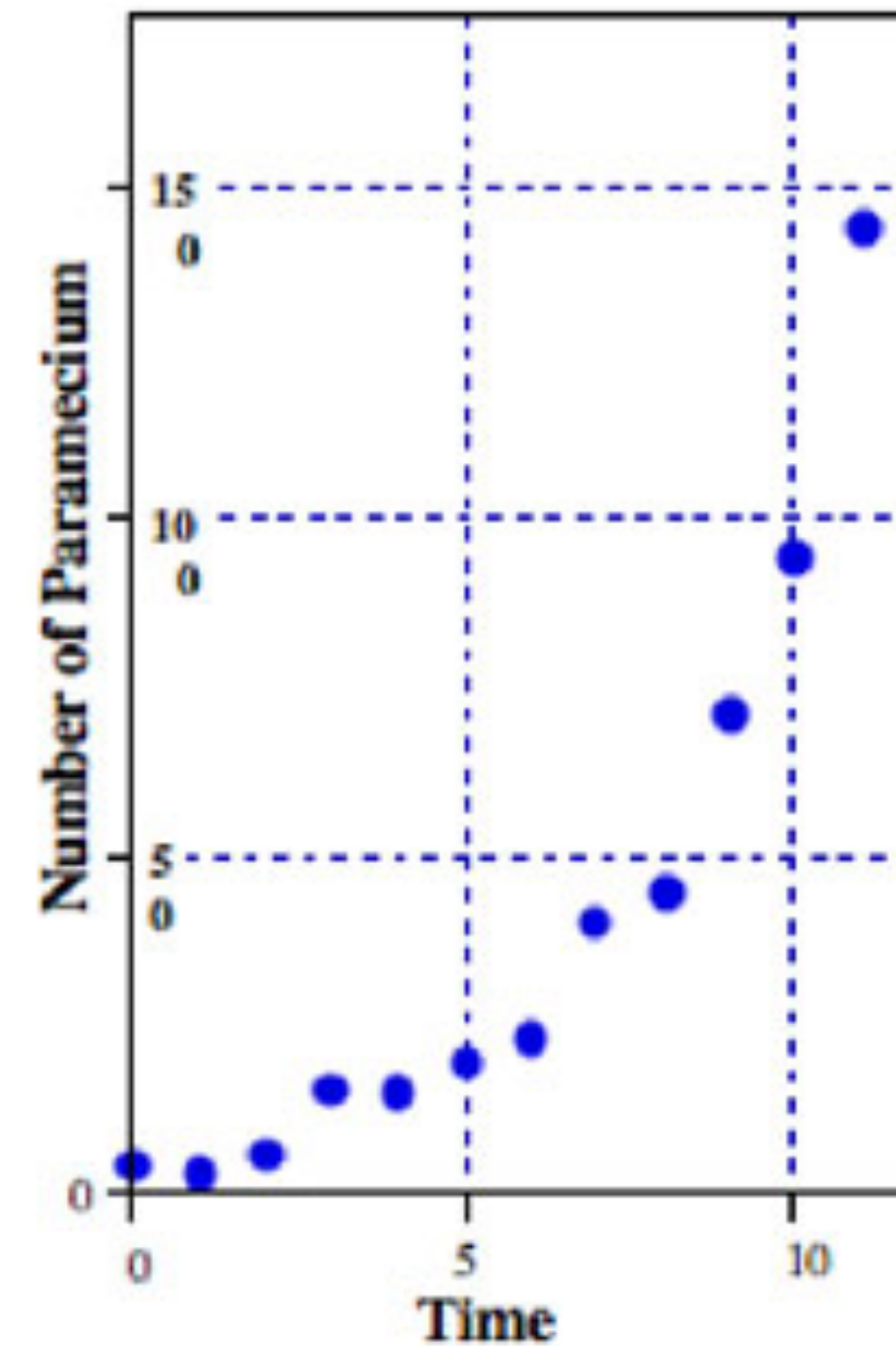
Intrinsic rate of increase

Idealised models help us understand population growth, BUT! they don't exist in Nature

Exponential growth

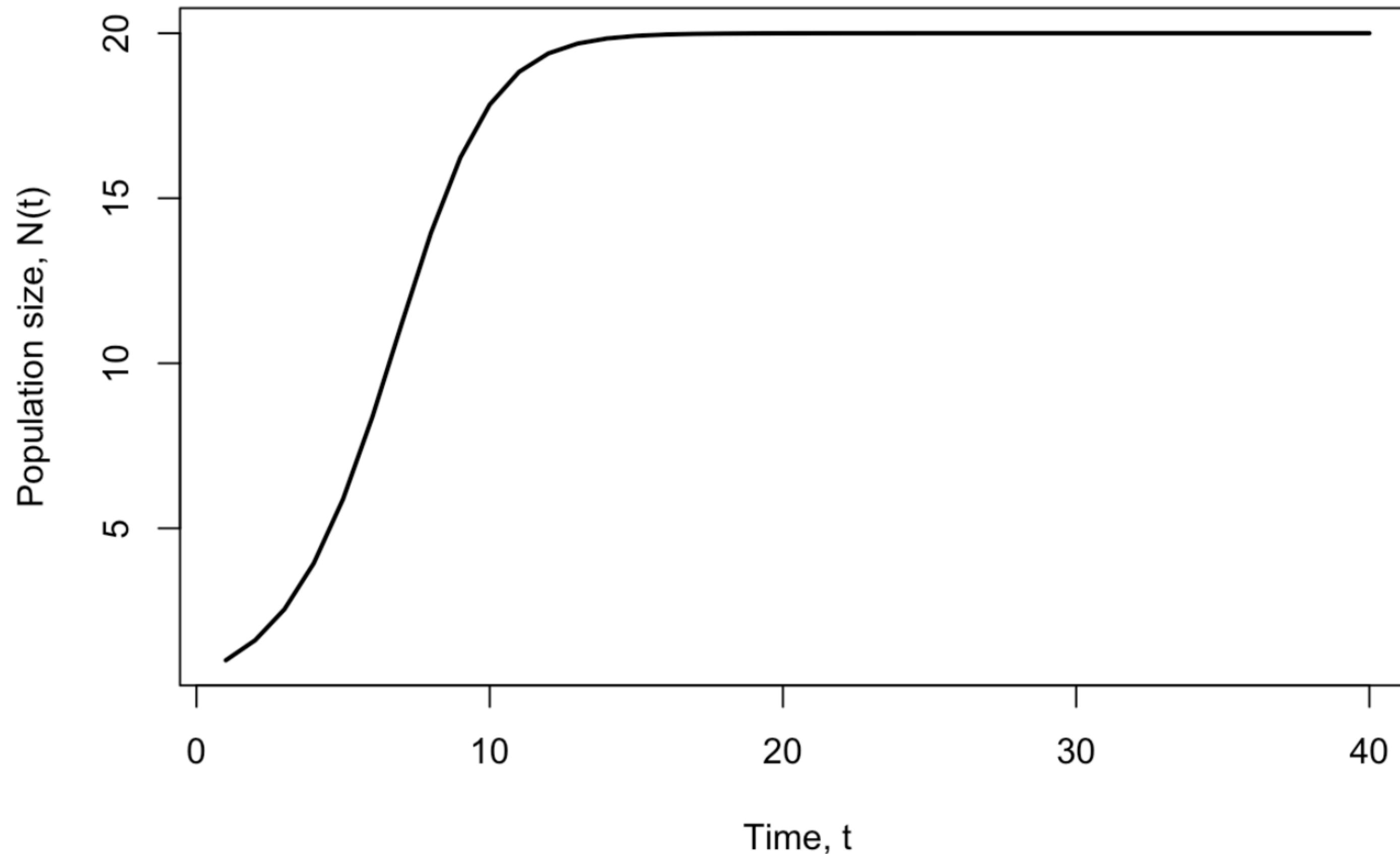


Modeled output



Empirical data

Density-dependent growth



Density dependence:

- Birth and death rates vary through time and depend on the density of individuals
- There is at least a limiting resource
- This generates competition between individuals

Density-dependent growth

The Ricker model

$$N_{t+1} = N_t * e^{r(1 - \frac{N_t}{K})}$$

Density-dependent growth rate

Intrinsic growth rate

Carrying capacity

The diagram shows the Ricker model equation $N_{t+1} = N_t * e^{r(1 - \frac{N_t}{K})}$. A red box highlights the exponential term $e^{r(1 - \frac{N_t}{K})}$, with a red arrow pointing to it from the text 'Density-dependent growth rate'. Inside this box, the parameter r is circled in green, with a green arrow pointing to it from the text 'Intrinsic growth rate'. Similarly, the term $\frac{N_t}{K}$ is circled in green, with a green arrow pointing to it from the text 'Carrying capacity'.

Species interactions

Interaction types

	-	+
-	- / - competition	- / + predation
+	+ / - predation	+ / + mutualism

Dynamics of two interacting species

Lotka (1910) - Autocatalytic chemical reactions

$$\frac{dx}{dt} = \alpha x - \beta xy$$

$$\frac{dy}{dt} = \delta xy - \gamma y$$

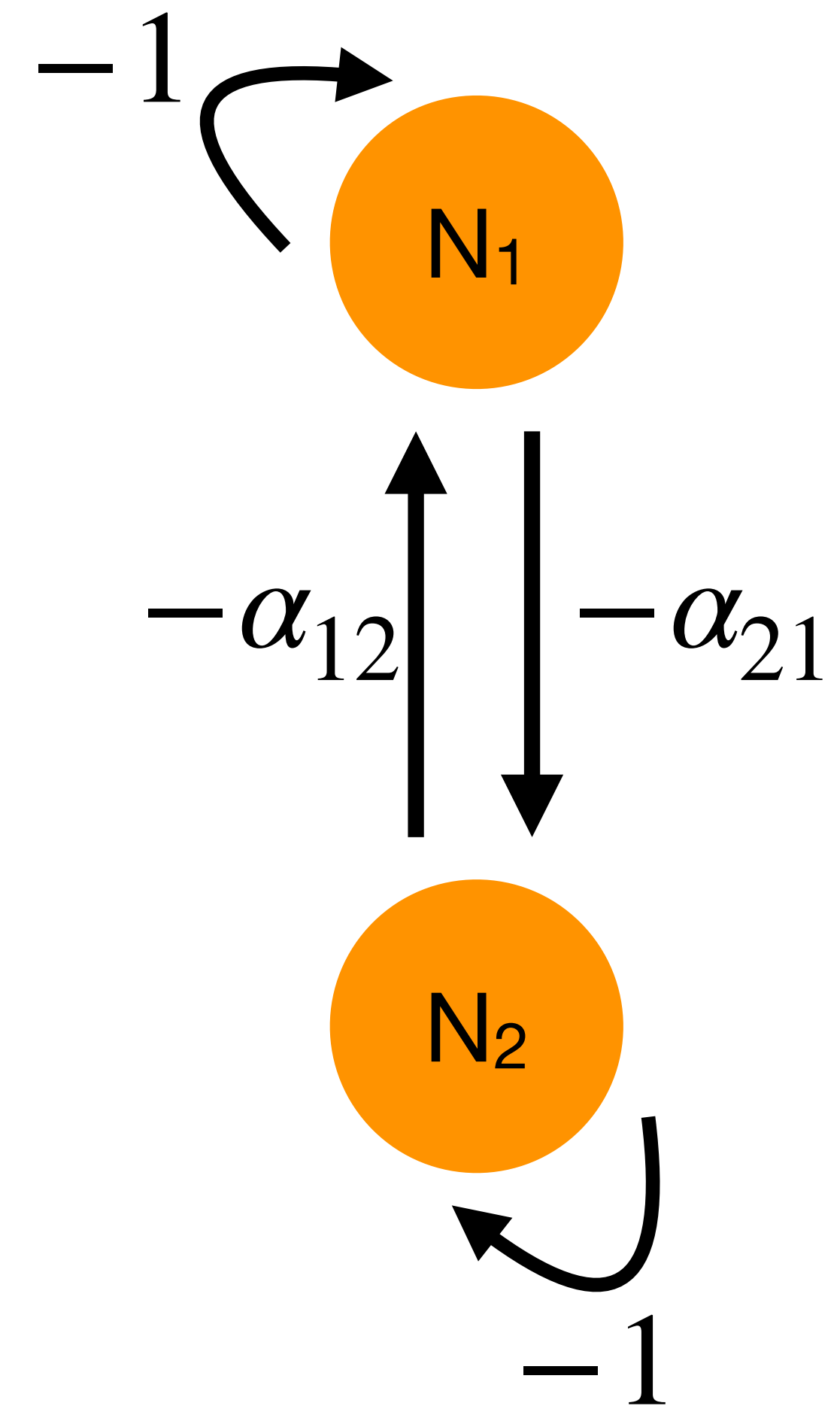
Volterra (1926) - Fisheries in the Adriatic Sea



Competitive Lotka-Volterra Model (with Density-Dependence)

$$\frac{dN_1}{dt} = r_1 N_1 \left(\frac{K_1 - N_1 - \alpha_{12} N_2}{K_1} \right)$$

$$\frac{dN_2}{dt} = r_2 N_2 \left(\frac{K_2 - N_2 - \alpha_{21} N_1}{K_2} \right)$$



Competitive Lotka-Volterra Model: finding equilibria

$$\frac{dN_i^*}{dt} = 0 = r_i N_i^* \left(\frac{K_i - N_i^* - \alpha_{ij} N_j^*}{K_i} \right)$$

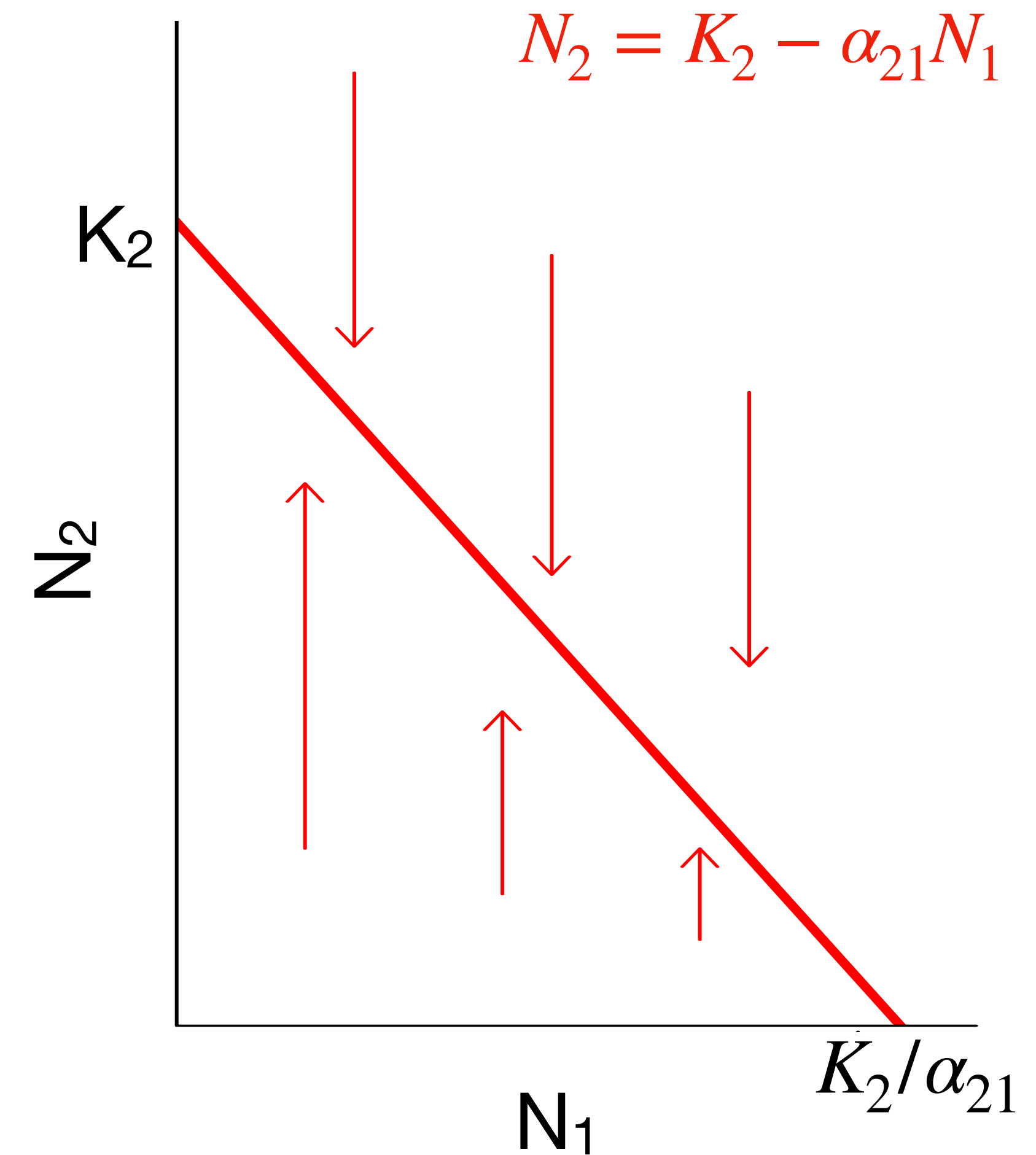
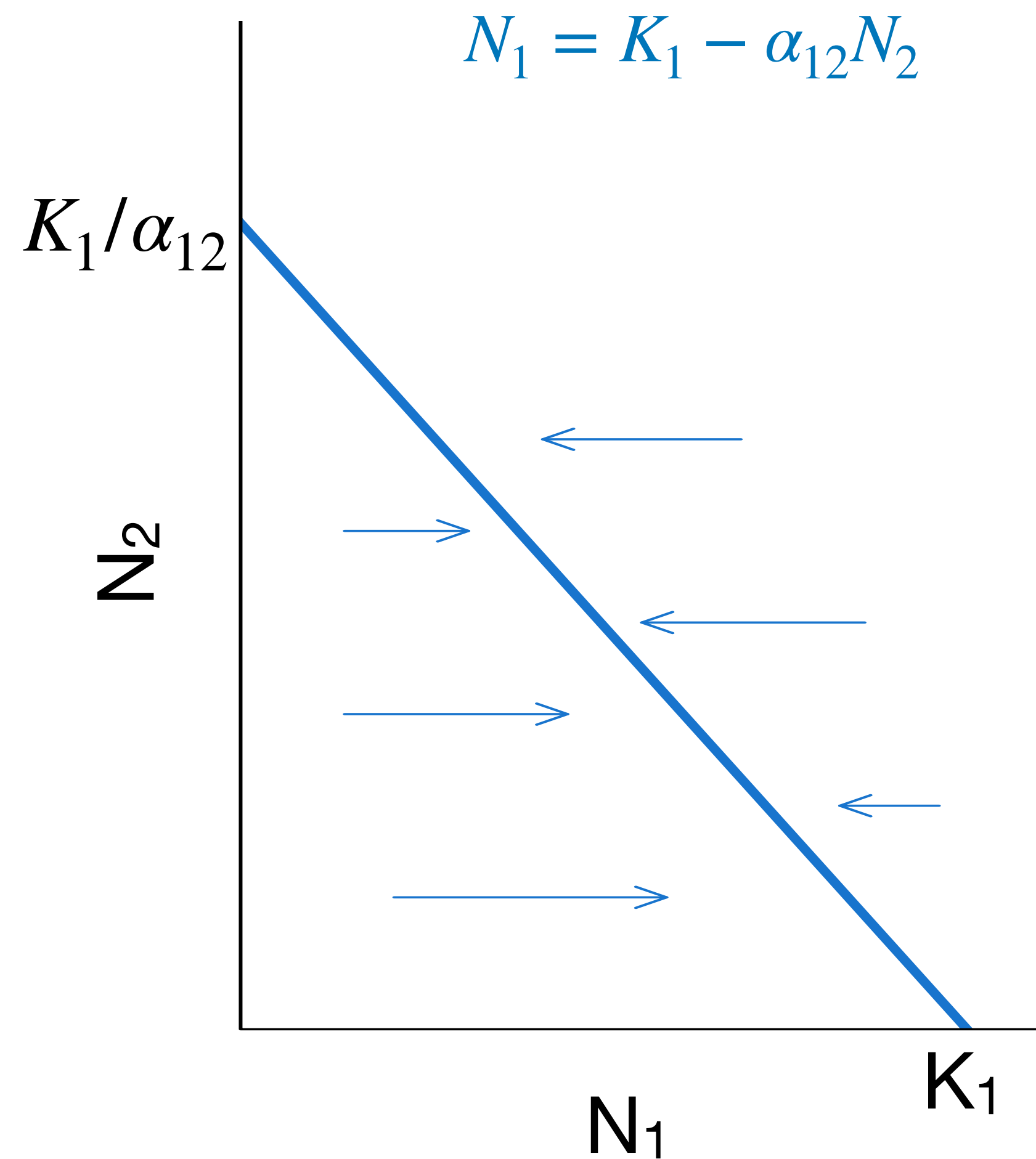
$$0 = K_1 - N_1^* - \alpha_{12} N_2^*$$

$$N_1 = K_1$$

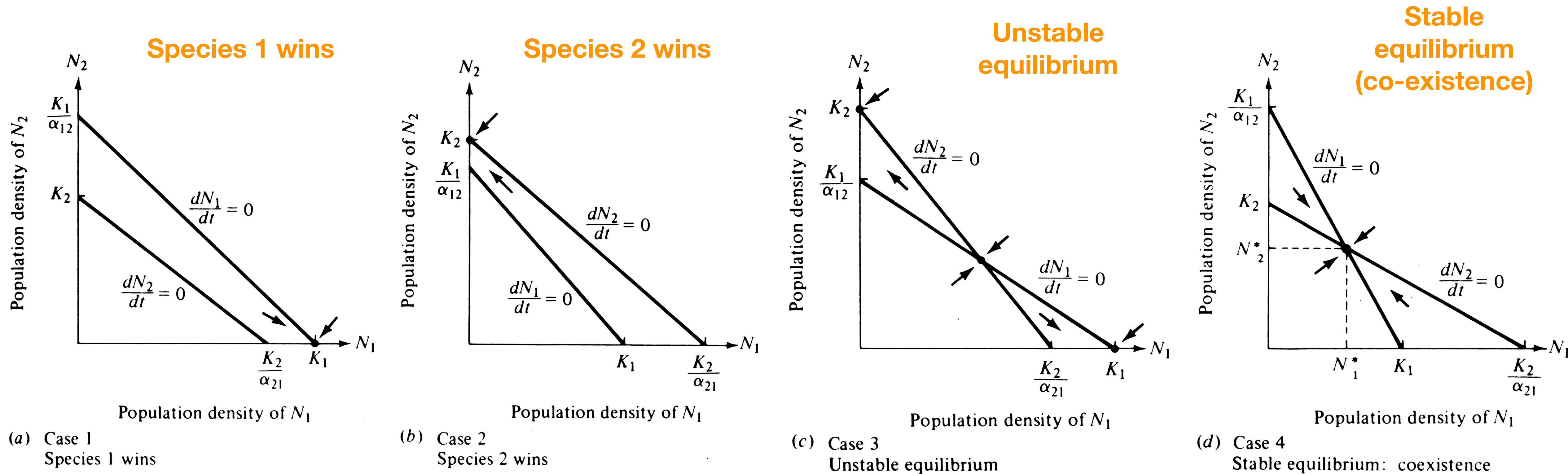
$$N_1 = K_1 - \alpha_{12} N_2 \quad \Rightarrow$$

$$N_2 = K_1 / \alpha_{12}$$

Competitive Lotka-Volterra Model: phase planes



Competitive Lotka-Volterra: phase plots



Constraint on stable coexistence

$$K_1 < \frac{K_2}{\alpha_{21}}$$

$$K_2 < \frac{K_1}{\alpha_{12}}$$

$$K_2 < \frac{K_2/\alpha_{21}}{\alpha_{12}}$$

$$1 < \frac{1}{\alpha_{21}\alpha_{12}}$$

$$K_2 < \frac{K_2}{\alpha_{21}\alpha_{12}}$$

$$1 > \alpha_{21}\alpha_{12}$$

Predator-prey Lotka-Volterra Equations

$$\frac{dN_2}{dt} = \alpha_{21}N_1N_2 - d_2N_2$$



Predator

$$\frac{dN_1}{dt} = r_1N_1 - \alpha_{12}N_1N_2$$



Prey



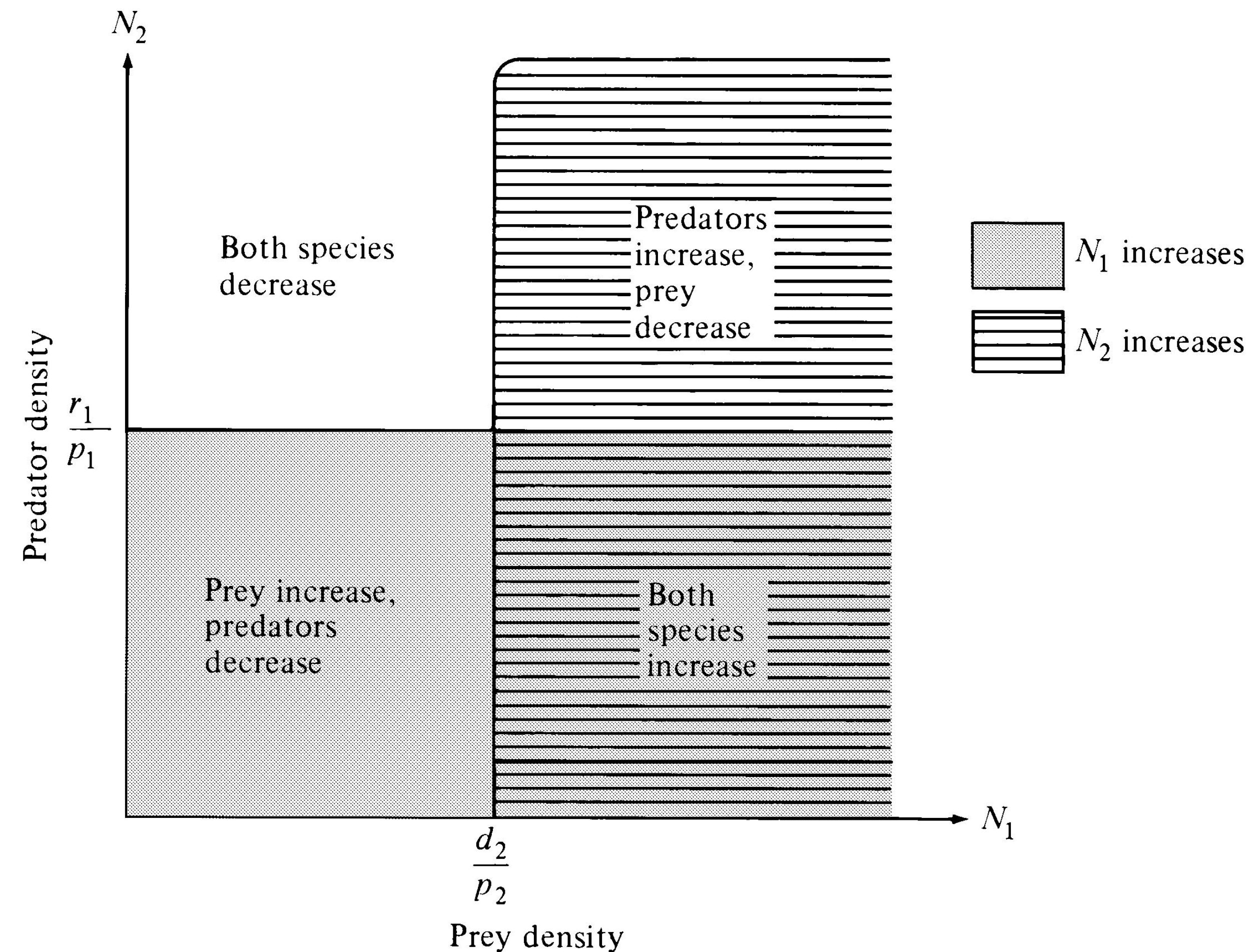
Predator-prey Lotka-Volterra Model

$$\frac{dN_2}{dt} = \alpha_{21}N_1N_2 - d_2N_2$$

Predator

Prey

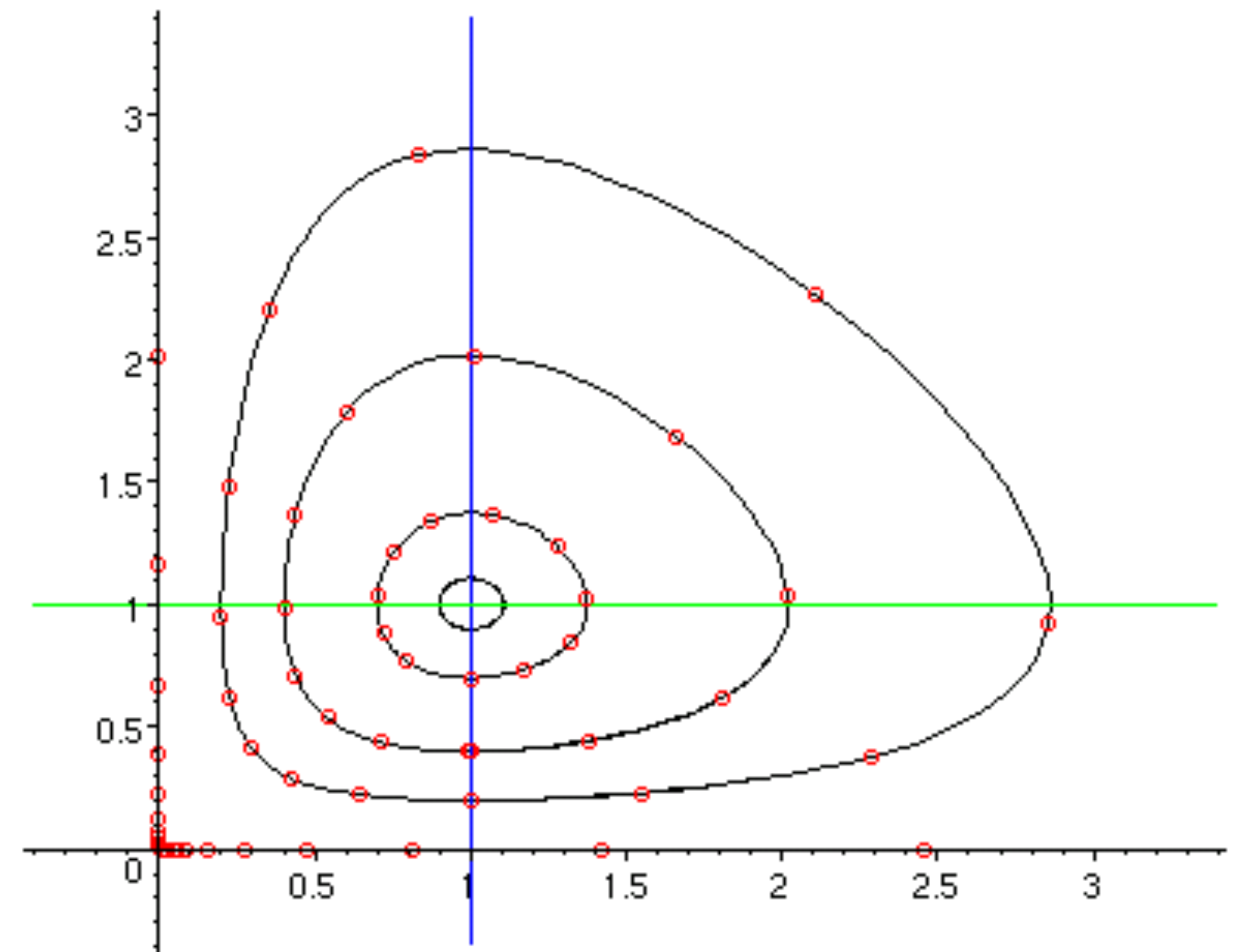
$$\frac{dN_1}{dt} = r_1N_1 - \alpha_{12}N_1N_2$$



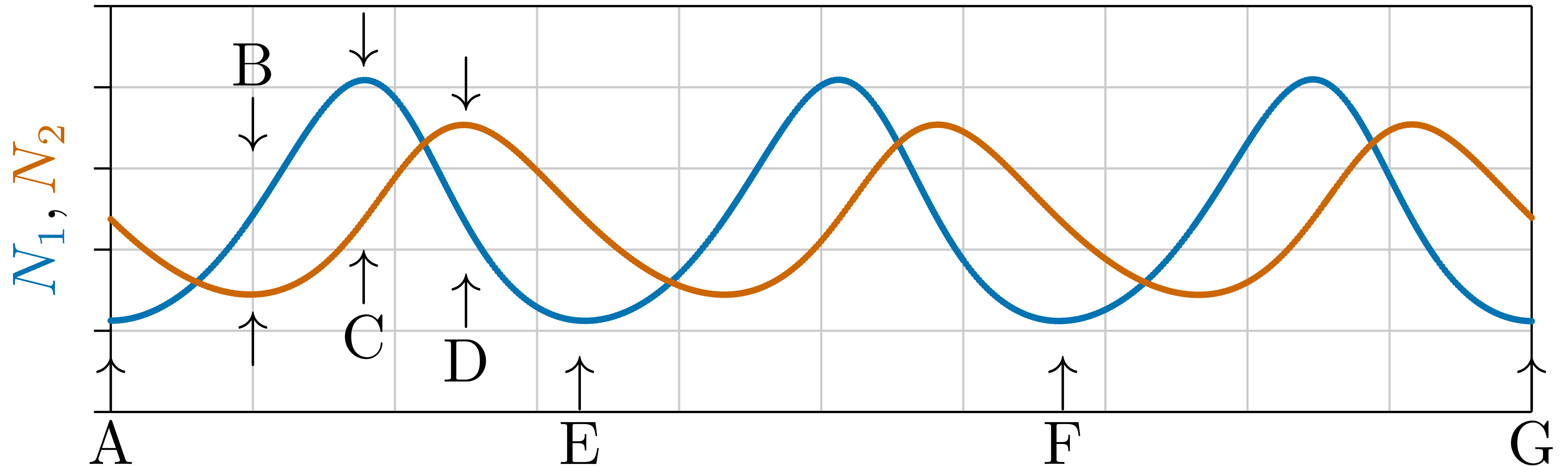
Predator-prey Lotka-Volterra Equations

$$\frac{dN_2}{dt} = \alpha_{21}N_1N_2 - d_2N_2$$

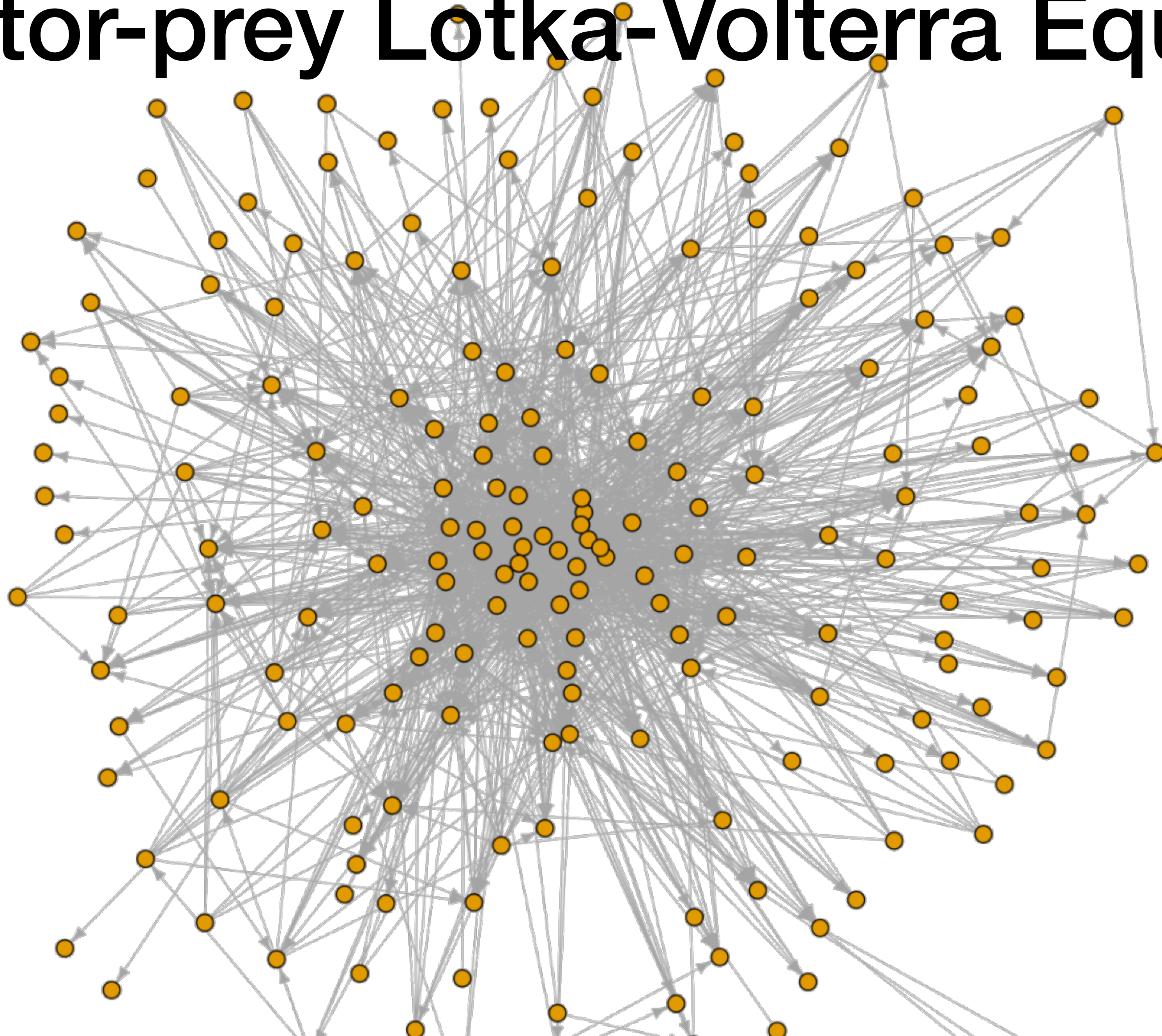
$$\frac{dN_1}{dt} = r_1N_1 - \alpha_{12}N_1N_2$$



Predator-prey Lotka-Volterra Equations



Predator-prey Lotka-Volterra Equations



Predator-prey Lotka-Volterra Equations

$$\frac{dN_2}{dt} = \alpha_{21}N_1N_2 - d_2N_2$$

Starting from a 2-species predator-prey system and extrapolating to many species...

$$\frac{dN_1}{dt} = r_1N_1 - \alpha_{12}N_1N_2$$

$$\frac{dN_i}{dt} = N_i \left(b_i + \sum_{j=1}^n \alpha_{ij}N_j \right), i = 1 \dots n \quad \xrightarrow{\text{Network in matrix form}} \quad \frac{dN}{dt} = N * (b + AN)$$

Methods for data simulation

Methods for data simulation

1. Difference equations
2. Ordinary differential equations
3. Individual-based models

Methods for data simulation

Difference equations

- Discrete time steps
- Calculate next abundance from previous

$$N_{t+1} = N_t + B_t + I_t - D_t - E_t$$

$$N_{t+1} = N_t * e^{r\left(1 - \frac{N_t}{K}\right)}$$

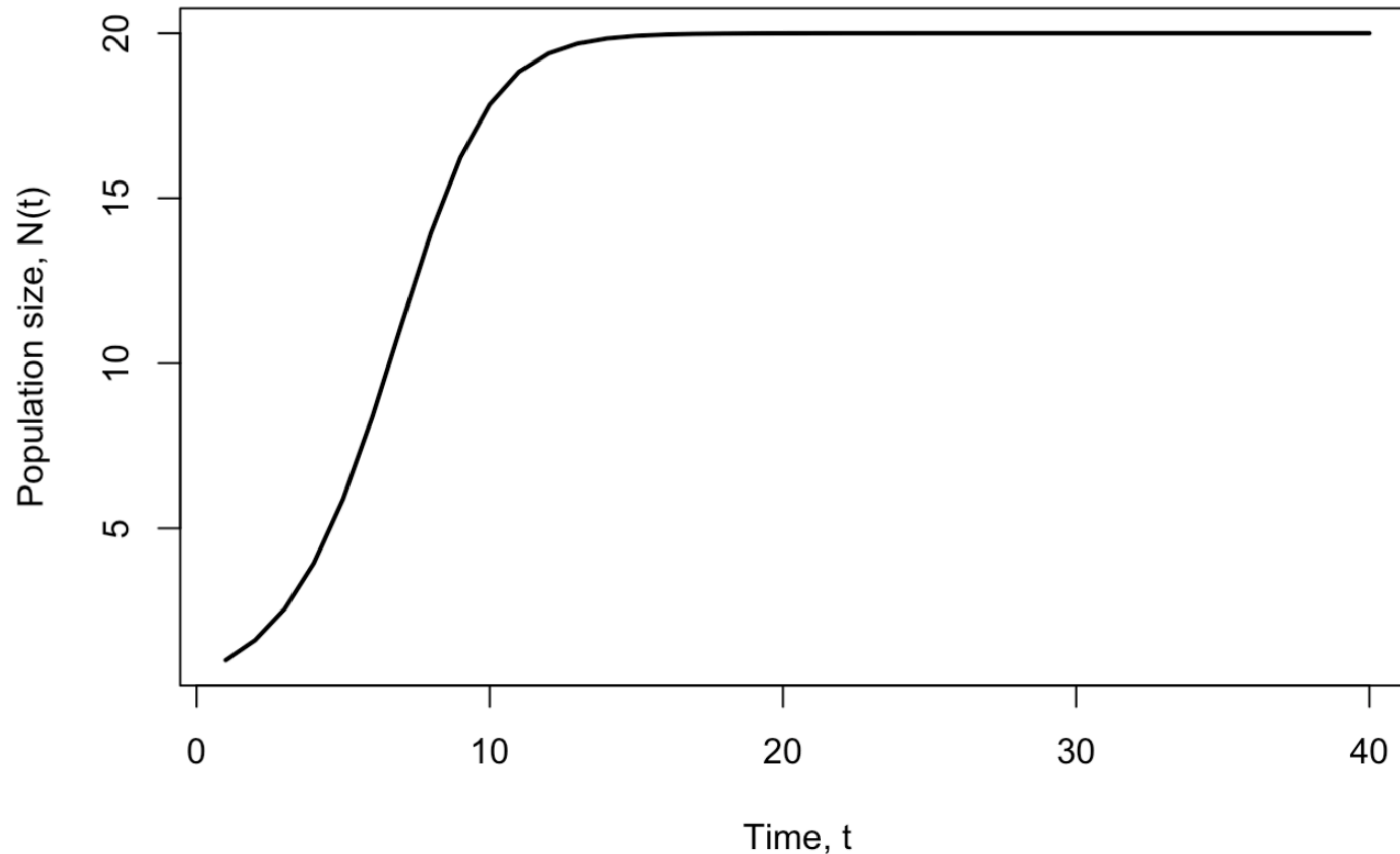
Developing the code

```
r <- 0.5      # Intrinsic Growth Rate
K <- 20       # Carrying Capacity
Tmax <- 40    # Maximum simulation time

# Pre-allocate population vector
N <- matrix(0, nrow = Tmax, ncol = 1)
N[1] <- 1 # Initialise population size

# Simulate Ricker population growth over time
for (t in 1:(Tmax-1)) {
  N[t+1] <- N[t]*exp(r*(1-N[t]/K))
}
```

Density-dependent growth



Density dependence:

- Birth and death rates vary through time and depend on the density of individuals
- There is at least a limiting resource
- This generates competition between individuals

Methods for data simulation

Ordinary differential equations (ODEs):

- Track the rate of change of some quantity through time
- Solved using tools from calculus
- Integrated using numerical integrators (what will use here)

$$\frac{dN}{dt} = N * (b + AN)$$

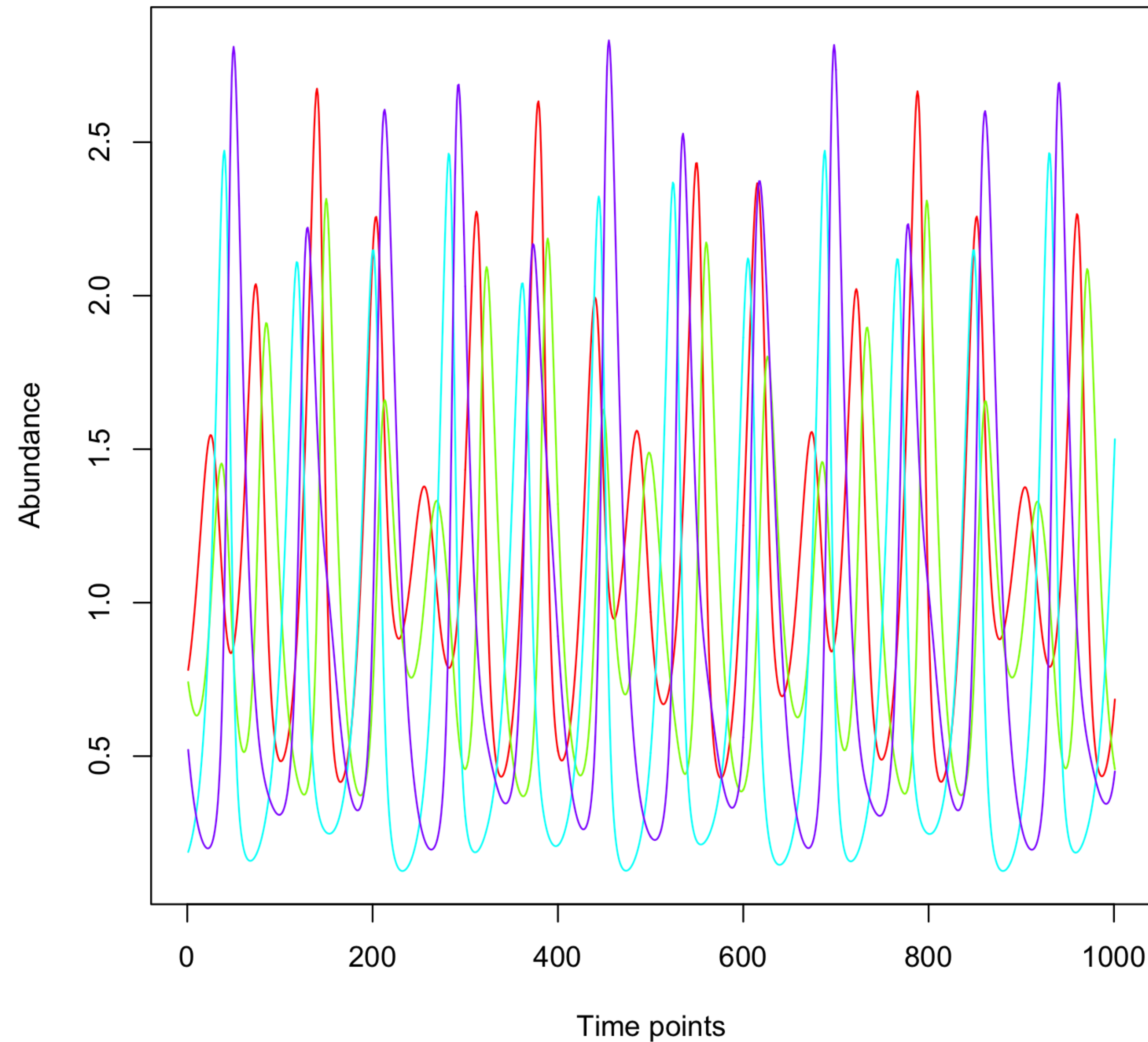
Developing the code

```
generalised.LV <- function(t, x, parameters){  
  with(as.list(c(x, parameters)), {  
    x[x < 10-8] <- 0 # to prevent numerical problems  
    dxdt <- x * (b + A %*% x)  
    list(dxdt)  
  })  
}
```

```
glv.out <- ode(y = N0, times = t.values, func =  
generalised.LV, parms = params)
```

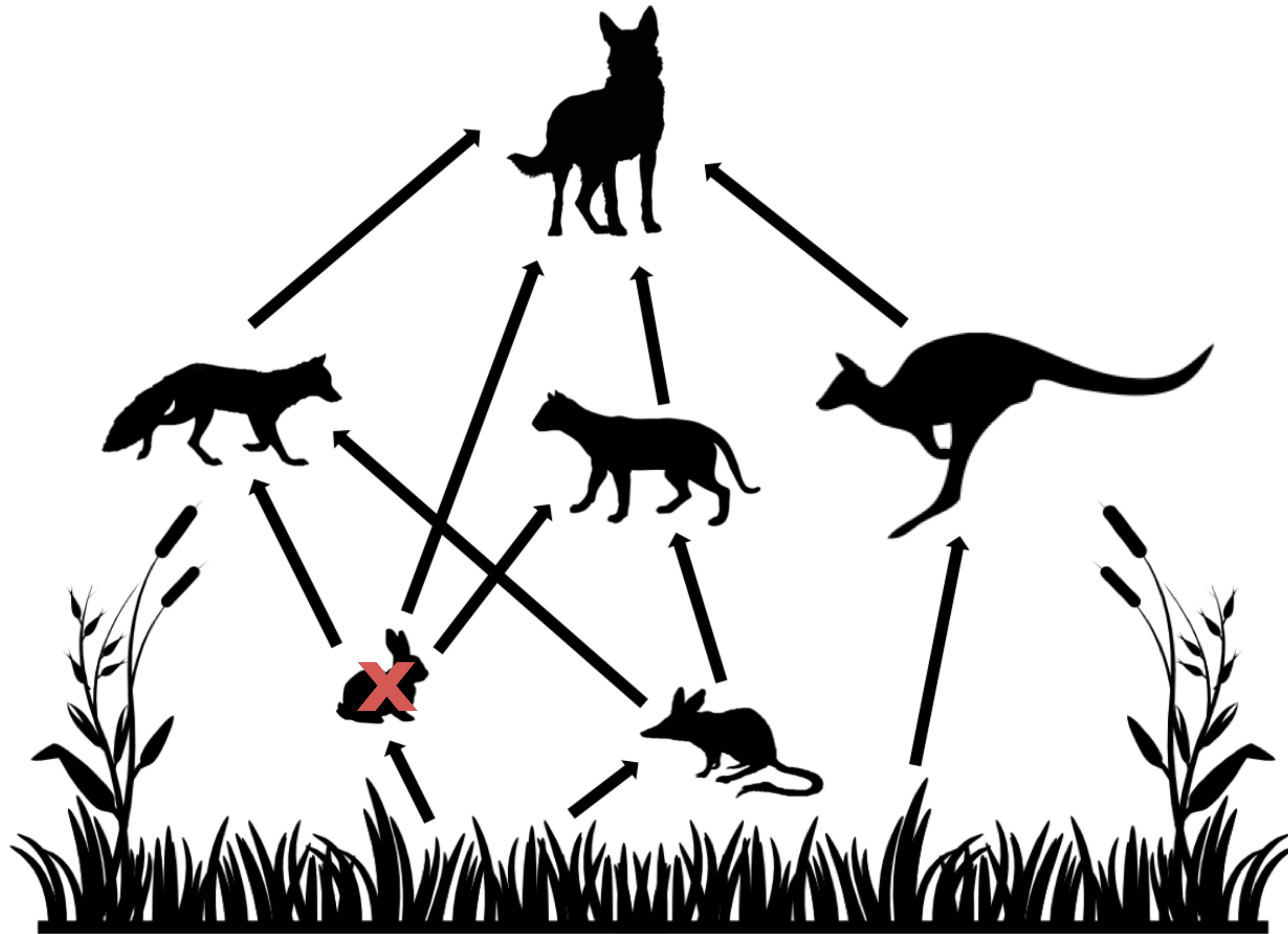
Predator-prey Continuous Equations

Community time series



Experimental protocols

Applied ecology



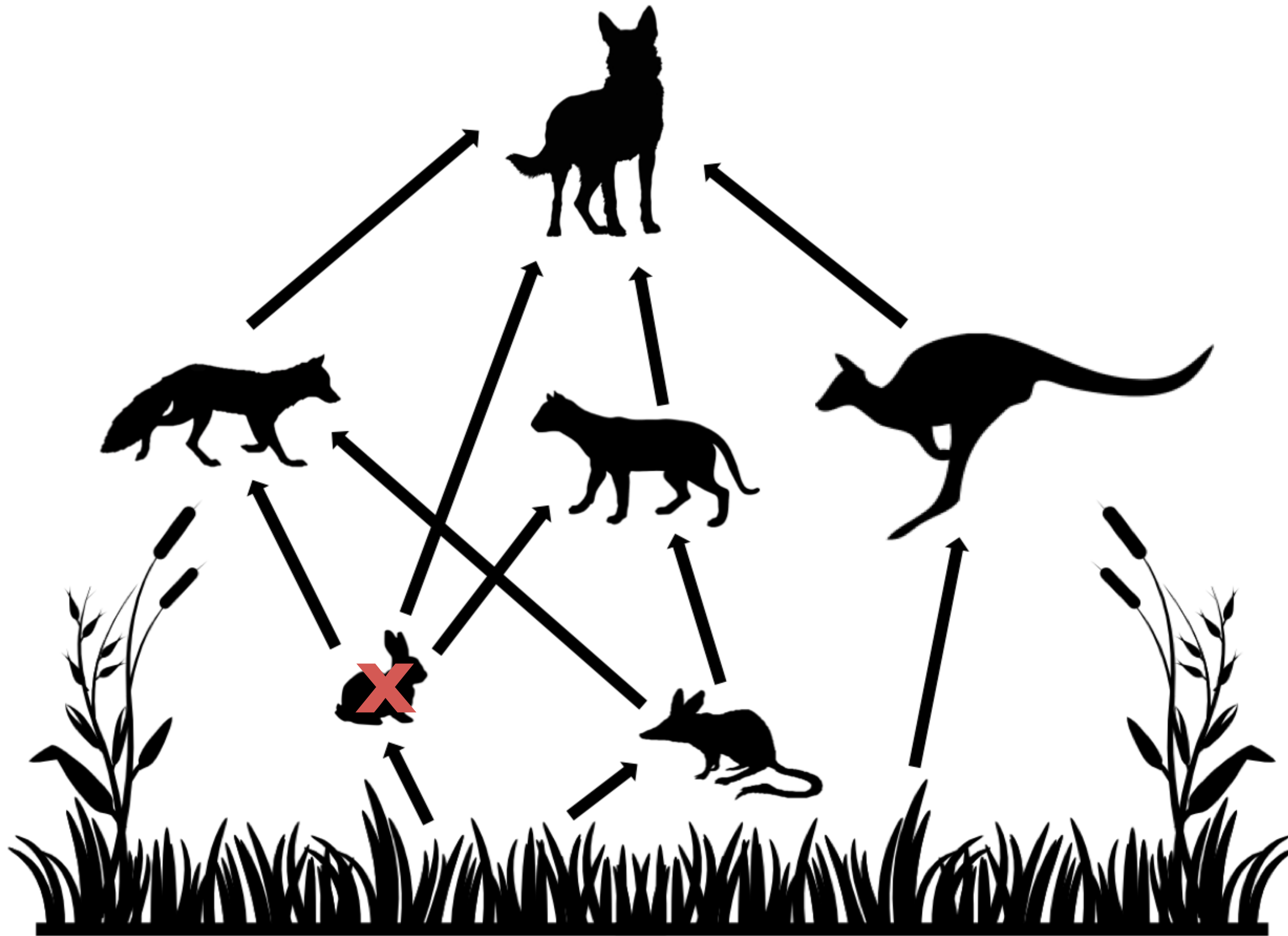
Resources

$$\Delta V = -55.12 - 0.0153V - 0.00056V^2 + 2.5Y$$

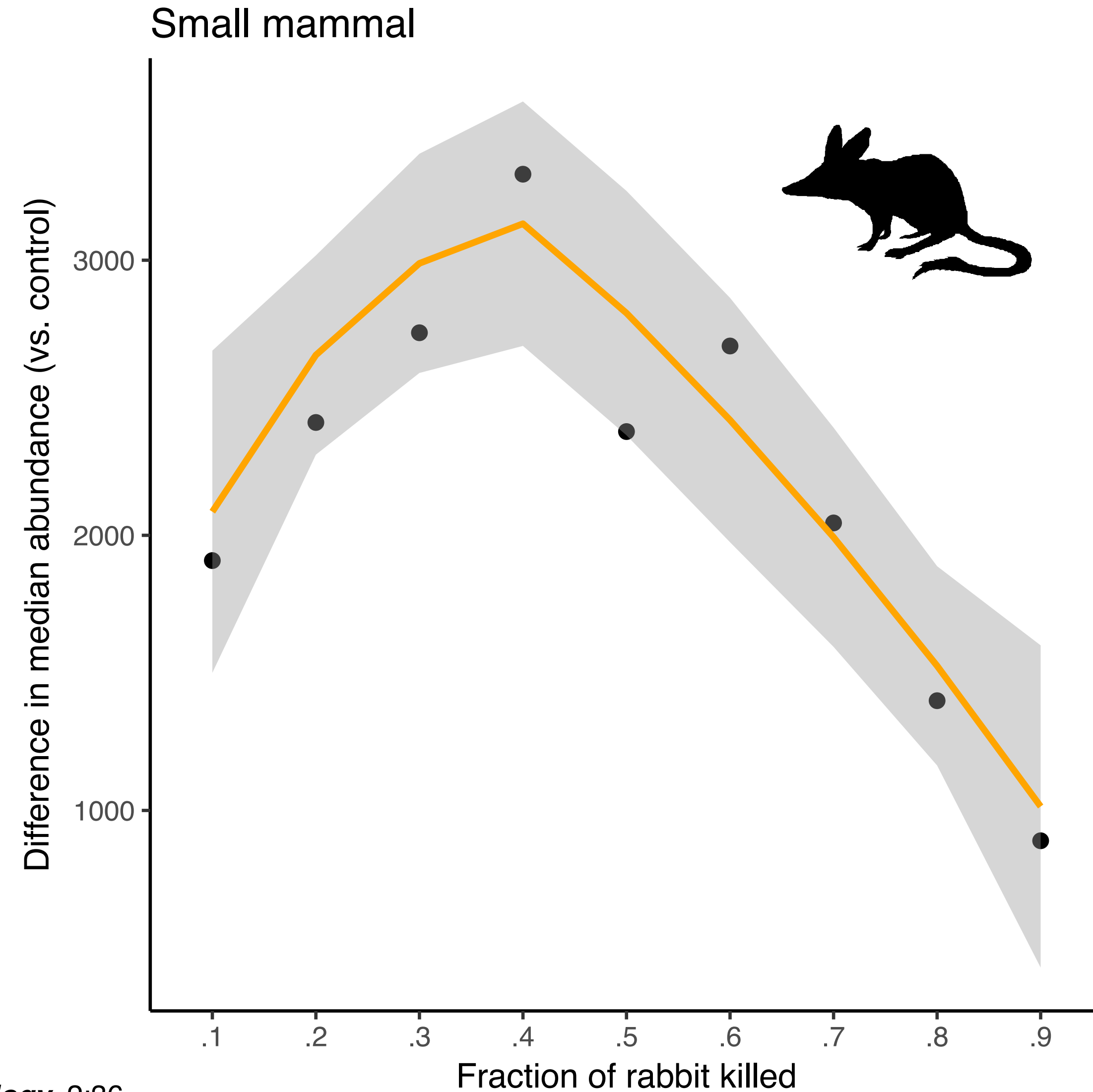
Consumers

$$r_{N,t} = -a_N + c_N \left[1 - \exp(-d_N V_{t-1}) \right] - P_{t-1}$$

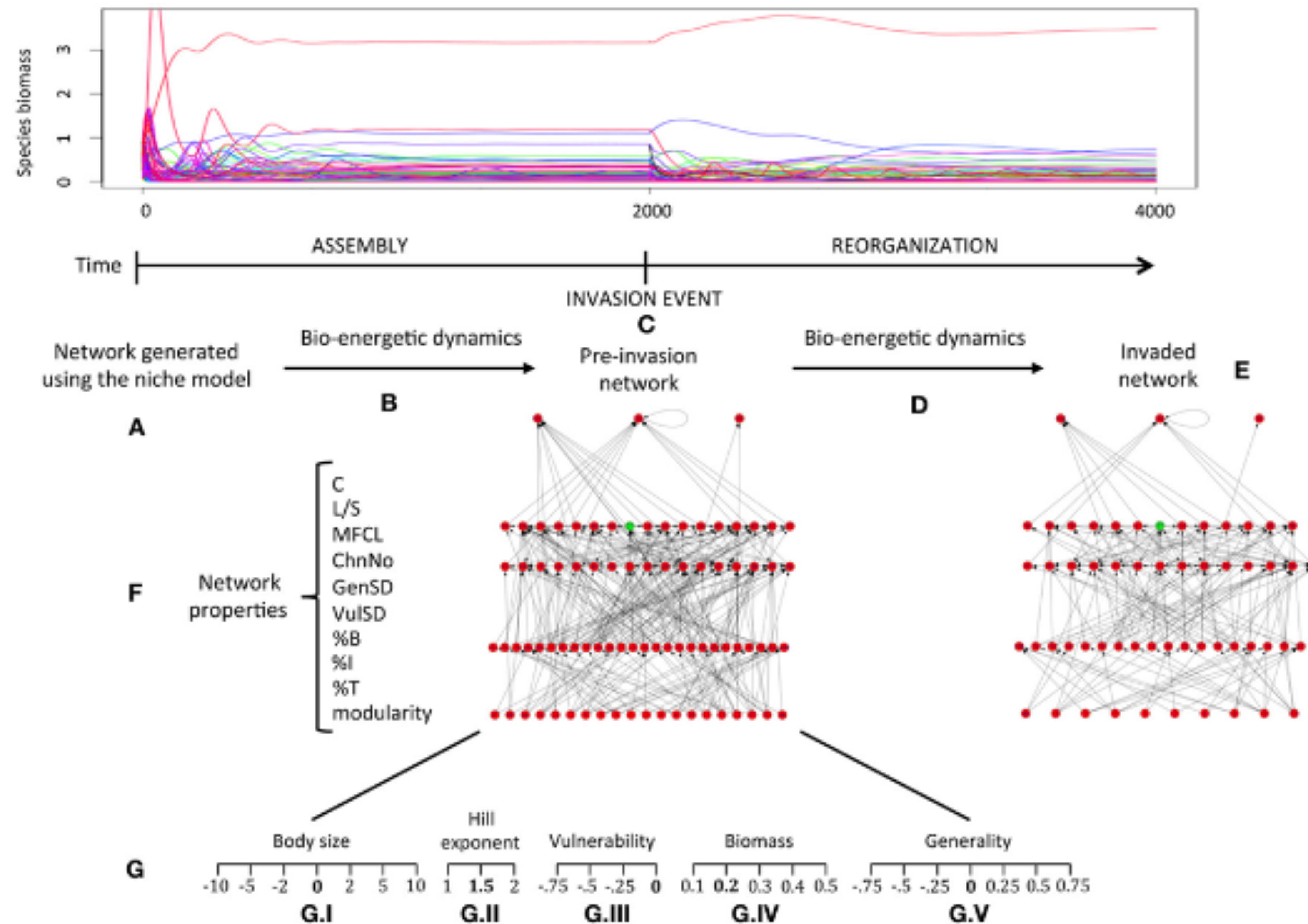
Applied ecology



Lurgi M et al. (2018). *Journal of Applied Ecology*, 2:36

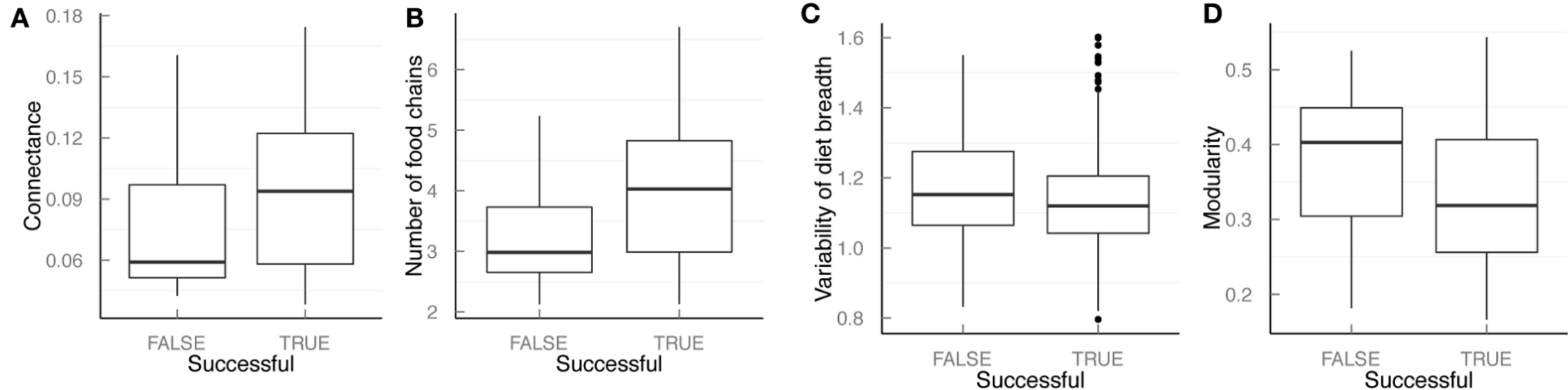


Applied ecology



Applied ecology

Network properties as predictors of invasion success



Knowledge recap

1. Population ecology - mechanisms of population change
2. Single species populations - exponential and density-dependent growth
3. Species interactions - Competition and predation
4. Lotka-Volterra model for network dynamics
5. Methods for simulating network dynamics - difference and differential equations
6. Experimental protocols to address ecological questions